125101 - Analytical chemistry 1 for engineers

Tutorial 5 – Linear regression

T5-1. In the Bradford protein determination, the color of the dye changes from brown to blue when it binds to the protein. In such a research, the absorbance at 595 nm was measured as a function of the mass of protein in a series of experiments, as shown in the table. Find the correlation of the protein mass and the absorbance at 595 nm by the least squares method. Then, calculate how much the protein mass when the absorbance at 595 nm is 0.947?

Protein (μg)	**Absorbance at 595 nm**
0.00	0.466
9.36	0.676
18.72	0.883
28.08	1.086
37.44	1.280

$$k = \frac{N\left(\sum x_i y_i\right) - \left(\sum x_i\right)\left(\sum y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

$$c = \frac{\left(\sum x_i^2\right)\left(\sum y_i\right) - \left(\sum x_i\right)\left(\sum x_i y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

	x_i	y_i	$x_i y_i$	x_i^2
1	0.00	0.466	0	0
2	9.36	0.676	6.327	87.61
3	18.72	0.883	16.530	350.44
4	28.08	1.086	30.495	788.49
5	34.77	1.280	47.923	1401.75
Sum	93.60	4.391	101.275	2628.29

$$k = \frac{5\times101.275 - 93.60\times4.391}{5\times2628.29 - 93.60^2} = 0.021773 = 0.0218$$

$$c = \frac{2628.29\times4.391 - 93.60\times101.275}{5\times2628.29 - 93.60^2} = 0.470605 = 0.471$$

So, the equation of the least squares straight line is

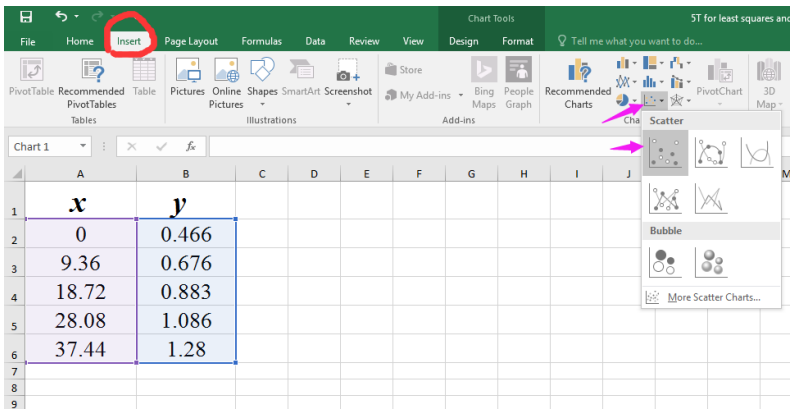
$$y = 0.0218x + 0.471$$

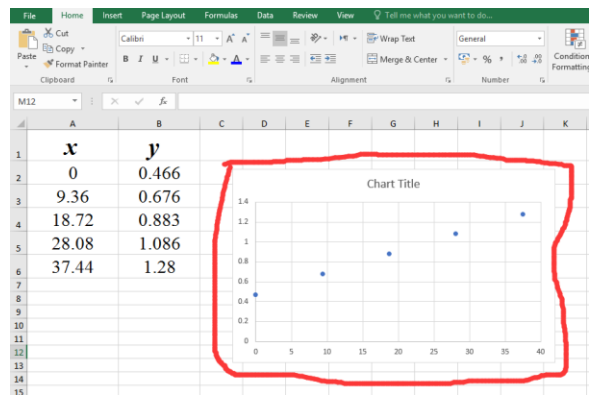
$$\text{when } y = 0.947,\ x = 21.83\ \mu g$$

How to use Excel to find the equation of least squares.

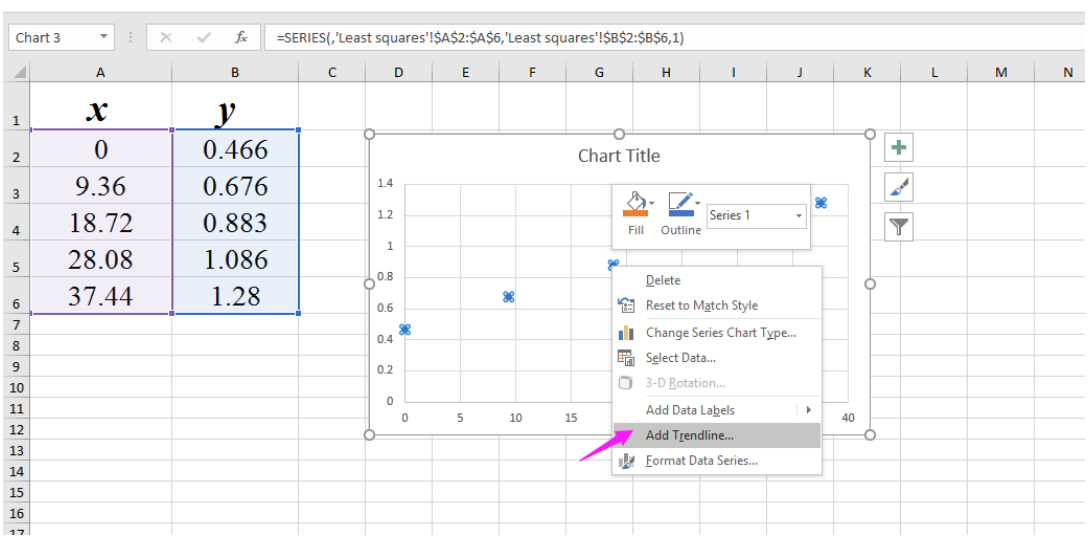
Attached is how to use spreadsheet of "*Least squares*" to find the answer.

1. Import the group of data in Excel of the values of *x* and *y*.
2. Choose the data, then click the bottom of "*Insert*" – "*Scatter*", choose it. You will get a chart.

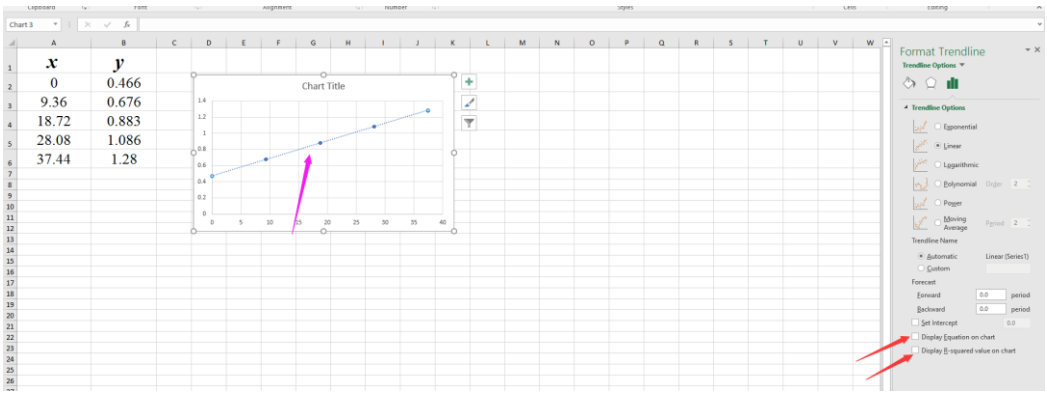


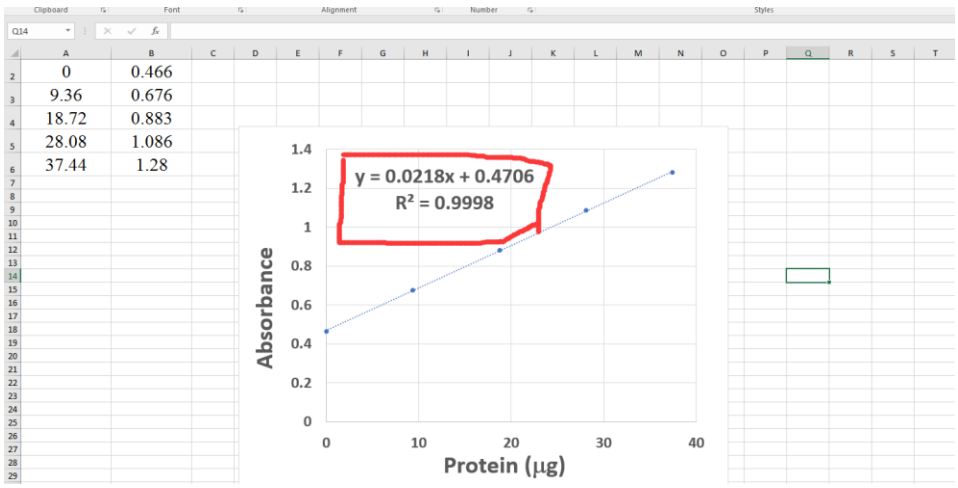


3. Choose the data point in the chart, then click the right bottom of mouse, "Add Trendline…" will appear, and choose it. The "best" straight line by least squares to fit the data would be done.



4. On the right of Excel, choose the "Display Equation on chart" and "Display R-squared value on chart", you will find the equation. The result will be the same with your calculation (y = 0.0218x + 0.0471).





T5-2. The measured current of an unknown sample of Ni^{2+} is 2.36 μA in an electrochemical analysis. When 0.500 mL of solution containing 0.0287 mol/L Ni^{2+} was added to 25.0 mL of unknown sample, the measured current is 3.79 μA. Find how much the initial concentration of Ni^{2+} in the unknown sample.

$$[Ni^{2+}]_f = [Ni^{2+}]_i \frac{V_o}{V} = [Ni^{2+}]_i \boxed{\frac{25.0\ mL}{25.5\ mL}}$$

Dilution factor

$$= 0.980_4\ [Ni^{2+}]_i$$

$$[S]_f = [S]_i \frac{V_s}{V} = 0.0287\ M\ \frac{0.500\ mL}{25.5\ mL}$$

$$= 0.000\ 562_7\ M$$

$$\frac{[Ni^{2+}]_i}{[Ni^{2+}]_f + [S]_f} = \frac{I_{[Ni^{2+}]}}{I_{([Ni^{2+}]+S)}}$$

$$\frac{[Ni^{2+}]_i}{0.980_4\ [Ni^{2+}]_i + 0.000\ 562_7\ M} = \frac{2.36\ \mu A}{3.79\ \mu A}$$

$$[Ni^{2+}]_i = 0.000899_{59}\ M = 9.00\times10^{-4}\ M$$

T5-3. Pb(II) was extracted from the dry river sediment with 25% HNO_3 at 35 °C for 1 hour. Then the extract with a volume of V_o = 4.60 mL. Then, the concentration of Pb(II) was measured electrochemically with a series of standard additions of 2.50 ppm Pb(II). Try to find the concentration of Pb(II) in the original extract volume.

Pb(II) added (mL)	**Signal (arbitrary units)**
0	1.10
0.025	1.66
0.050	2.20
0.075	2.81

First, we need to calculate the values of *x*-*y*. For successive standard addition to a single solution of unknown, the value of $\boldsymbol{y}$ is $\boldsymbol{I}_{(\boldsymbol{X}+\boldsymbol{S})}(\frac{\boldsymbol{V}}{\boldsymbol{V_o}})$and $\boldsymbol{x}$ is $[\boldsymbol{S}]_{\boldsymbol{i}}(\frac{\boldsymbol{V_s}}{\boldsymbol{V_o}})$.

Pb(II) Added (mL)	$\boldsymbol{x} - [\boldsymbol{S}]_{\boldsymbol{i}}(\frac{\boldsymbol{V_s}}{\boldsymbol{V_o}})]$ ppm	Signal	$\boldsymbol{y} - \text{Signal}\times\frac{\text{V}}{\text{V}_\text{O}}$
0	0	1.10	$1.10\times\frac{4.60\text{ mL}}{4.60\ mL}$
0.025	$2.50\times\frac{0.025}{4.60\ mL}$	1.66	$1.66\times\frac{(4.60+0.025)\text{ mL}}{4.60\ mL}$
0.050	$2.50\times\frac{0.050}{4.60\ mL}$	2.20	$2.20\times\frac{(4.60+0.050)\text{ mL}}{4.60\ mL}$
0.075	$2.50\times\frac{0.075}{4.60\ mL}$	2.81	$2.81\times\frac{(4.60+0.075)\text{ mL}}{4.60\ mL}$

$\boldsymbol{x} - [\boldsymbol{S}]_{\boldsymbol{i}}(\frac{\boldsymbol{V_s}}{\boldsymbol{V_o}})]$/ppm	$\boldsymbol{y} - \text{Signal}\times\frac{\text{V}}{\text{V}_\text{O}}$
0	1.10
0.013_6	1.66_9
0.027_1	2.22_4
0.040_8	2.85_6

According to the method of least squares (or you can do it on Excel), you can get the function as: $y = 42.851x + 1.0892$, with $R^2 = 0.9993$

Set $y = 0$, the x-intercept is 0.02542

For a series of standard additions to a single solution of unknown, the total volume is **NOT** constant, the x-intercept is the concentration of analyte in sample.

So, Pb(II) in the extract is 0.02542 ppm.

Considering the significant figures, it is 0.0254 ppm Pb(II) in the extract.

T5-4. The light-sensitive chemicals on photographic film are silver halides. Thus, silver is often extracted from photographic waste for commercial reclamation. A sample of photographic waste containing an unknown amount of silver was determined by standard addition with atomic absorption spectroscopy.

Different volumes of standard $AgNO_3$ solution with the concentration of 0.50 mg/mL were added into seven volumetric flasks containing 10 mL of photographic waste. All the flasks were filled with water to 100 mL. The atomic absorption signals results for each of solution are listed in the table. How much the concentration of Ag^+ in the original solution of the photographic waste sample?

Added standard solution of $AgNO_3$ (mL)	**Absorbance (arbitrary units)**
0	0.32
1	0.41
2	0.52
3	0.60
4	0.70
5	0.77
6	0.89

For multiple solution with constant total volume, the following equation can be used,

$$I_{(X+S)} = \frac{I_X}{[X]_i}[X]_f + \frac{I_X}{[X]_i}[S]_f$$

The value of y is $\boldsymbol{I}_{(X+S)}$ and x is $[\boldsymbol{S}]_f$.

x-$[\boldsymbol{S}]_f$ (μg/mL)	y- $\boldsymbol{I}_{(X+S)}$
0	0.32
5	0.41
10	0.52
15	0.60
20	0.70
25	0.77
30	0.89

According to the method of least squares (or you can do it on Excel), you can get the function as,

$$y = 0.0186x + 0.3218, \text{ with } R^2 = 0.9976$$

Set $y = 0$, the x-intercept is 17.30.

And the x-intercept is the final concentration of $[X]_f$ in the flask. Compute the $[X]_i$ through the dilution factor, i.e. multiply by a factor of V/V_o.

So, the $[X]_f$ = x-intercept = 17.30 μg/mL

$$[X]_i = [X]_f \times \frac{V}{Vo} = 17.30 \times \frac{100\ mL}{10\ mL}\ \mu g/mL = 0.1730\ mg/mL$$

Considering the significant figures,

Therefore, 0.17 mg/mL Ag^+ in the original solution of the photographic waste sample.

Well, if you spike your sample with several known concentrations of analyte, you can only only plot the assay response (such as absorbance) versus added concentration of analyte. So, you'll still have a response as you cross the y-intercept with no added analyte since your sample hopefully contained some to start with.

The negative x-intercept is essentially how much you'd have to remove to get no response (aka: your starting concentration). It's the origin on a calibration curve, just the x-axis has been shifted since it's added concentration, not total concentration.

If you prefer to think in math, the line itself is y = mx + b. Since we want the x-intercept, y=0.

0= mx + b

$-b = mx$

$x = -b/m$

x is your x-intercept, b is your y-intercept in units of (assay response) and m is your slope in units of (assay response)/[concentration]. So the x-intercept is really just the y-intercept (response of your unspiked sample) divided by the slope (which is the same as the slope of a standard curve).

x = (response)/((response)/[concentration]) = [concentration]